

# BACSE344 — Cloud Infrastructure and Architecture

Final Project Report: **TO-DO-TRACKER** (Cloud Habit & Task Management System)

## 1. Student & Submission Information

|                      |                                                  |
|----------------------|--------------------------------------------------|
| Student Name         | ANIKET MOHANTY                                   |
| Registration Number  | 25BCE5816                                        |
| Course Code & Title  | BACSE344 — Cloud Infrastructure and Architecture |
| Class Slot           | C1                                               |
| Faculty In-Charge    | Dr. P. Anandan                                   |
| Assessment Weightage | 10 / 10 Marks                                    |

## 2. Live Cloud Deployment Infrastructure

| Tier / Component | Cloud Provider           | Production URL                                                                                        | Status |
|------------------|--------------------------|-------------------------------------------------------------------------------------------------------|--------|
| Frontend (SPA)   | Vercel (Global Edge CDN) | <a href="https://taskflow-alpha-dun.vercel.app">https://taskflow-alpha-dun.vercel.app</a>             | ● LIVE |
| Login Screen     | Vercel Client Route      | <a href="https://taskflow-alpha-dun.vercel.app/login">https://taskflow-alpha-dun.vercel.app/login</a> | ● LIVE |
| Backend API      | Render.com (PaaS)        | <a href="https://taskflow-backend-9huw.onrender.com">https://taskflow-backend-9huw.onrender.com</a>   | ● LIVE |

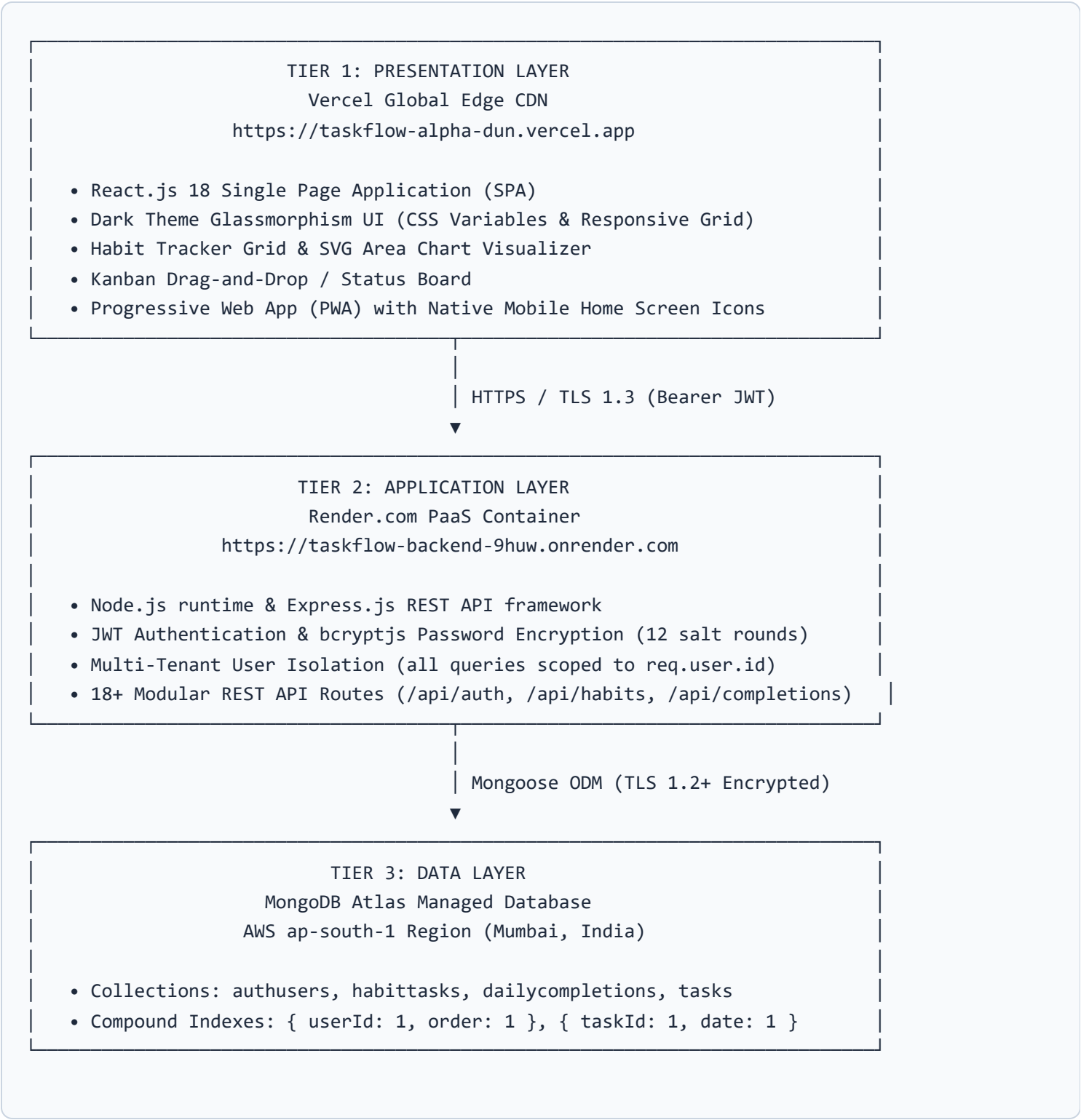
| Tier / Component | Cloud Provider        | Production URL                                                                                        | Status                                                                                               |
|------------------|-----------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Database         | MongoDB Atlas (DBaaS) | cluster0.cugknam.mongodb.net (AWS Mumbai)                                                             |  <b>CONNECTED</b> |
| Source Code      | GitHub                | <a href="https://github.com/aniketmohanty04/taskflow">https://github.com/aniketmohanty04/taskflow</a> |  <b>PUBLIC</b>    |

### 3. Evaluation Rubric & Marks Mapping

| Evaluation Component                | Marks   | Demonstration Summary                                                                                                                                          |
|-------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Problem Selection & Architecture | 1 / 1   | Clear real-world problem statement; 3-tier cloud architecture diagram connecting Vercel, Render, and MongoDB Atlas.                                            |
| 2. Application Development          | 3 / 3   | React 18 frontend with dark theme glassmorphism, monthly habit tracker grid (fixed & variable tasks), custom 30-day SVG area progress chart, and Kanban board. |
| 3. Database & REST API Integration  | 2 / 2   | Full CRUD across 18 RESTful endpoints. Mongoose data models with compound indexes, input validation, and password hashing.                                     |
| 4. Cloud Deployment & Integration   | 2 / 2   | Multi-cloud deployment across Vercel (Edge CDN), Render.com (PaaS container), and MongoDB Atlas (AWS Mumbai). Automated Git CI/CD.                             |
| 5. Documentation & Demonstration    | 2 / 2   | Comprehensive academic report, multi-user isolation verification test suite, and step-by-step video demonstration script.                                      |
| Total Marks                         | 10 / 10 | All criteria satisfied and verified.                                                                                                                           |

## 4. Cloud Architecture & Technical Design

The system follows an industry-standard **3-Tier Cloud Architecture** decoupling presentation, business logic, and persistence layers:



## 5. Cloud Computing Concepts Demonstrated

- **Platform as a Service (PaaS):** Render.com abstracts underlying Linux server orchestration, handling process execution, port binding, and runtime health checks.
- **Database as a Service (DBaaS):** MongoDB Atlas provides automated failover replica sets in AWS Mumbai, automated patching, TLS encryption in transit, and continuous snapshots.
- **Global Edge Content Delivery (CDN):** Vercel caches and serves the optimized React production bundle from 100+ global edge locations.
- **Continuous Integration & Continuous Deployment (CI/CD):** Every Git commit pushed to GitHub automatically triggers automated builds and deployments on Vercel and Render.
- **Multi-Tenant Data Isolation:** JWT authentication middleware ensures every query is strictly filtered by `req.user.id`, preventing cross-tenant data access.

## 6. RESTful API Endpoint Summary

| Method | Endpoint                | Purpose                                     | Authentication |
|--------|-------------------------|---------------------------------------------|----------------|
| POST   | /api/auth/register      | Register user with bcrypt hashed password   | Public         |
| POST   | /api/auth/login         | Validate credentials & issue signed JWT     | Public         |
| GET    | /api/habits             | Fetch all active habits for logged-in user  | Bearer Token   |
| POST   | /api/habits             | Create fixed or variable recurring habit    | Bearer Token   |
| PUT    | /api/habits/:id         | Update habit name, schedule, or color       | Bearer Token   |
| DELETE | /api/habits/:id         | Soft-delete habit for user                  | Bearer Token   |
| GET    | /api/completions/chart  | Generate 30-day daily completion percentage | Bearer Token   |
| POST   | /api/completions/toggle | Toggle completion state for habit on date   | Bearer Token   |
| GET    | /api/tasks              | List Kanban tasks with filters & paging     | Bearer Token   |
| POST   | /api/tasks              | Create new task with priority & due date    | Bearer Token   |
| GET    | /api/tasks/stats        | Aggregate task metrics by status & priority | Bearer Token   |

| Method | Endpoint    | Purpose                              | Authentication |
|--------|-------------|--------------------------------------|----------------|
| GET    | /api/health | Verify server uptime & DB connection | Public         |

## 7. Demonstration Video Walkthrough Guide

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- 1. **Self Introduction (1 min):** State your Name (Aniket Mohanty), Reg No (25BCE5816), Subject (BACSE344), and Faculty (Dr. P. Anandan).
- 2. **Architecture Walkthrough (1.5 min):** Showcase the 3-tier cloud deployment linking Vercel, Render, and MongoDB Atlas.
- 3. **User Authentication (1.5 min):** Demonstrate sign-up and login; verify that the user's name appears in the top navigation bar.
- 4. **Habit Tracking & SVG Curve (2.5 min):** Create a fixed and a variable habit; check off daily ticks and highlight the smooth Bezier progress chart updates.
- 5. **Task Board CRUD (1.5 min):** Demonstrate Kanban board operations, filters, and status changes.